# Descriptive Statistics and Probability for Business

Step-by-Step Solution Set

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# Hello, I am Jovan Samke!

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## My Story

- I am 24-year old guy from Serbia, and I went to **the best math high school in Europe.**
- I did my BSc in Mathematics, had **a 93% GPA**, and stacked up a bunch of awards.
- Right now, I am doing my MSc in Statistics and Data Science at KU Leuven, and **I am the only one who scored a full scholarship**.
- I have done hundreds of tutoring sessions, helping dozens of BBA students succeed, **nearly all of them passed their exams.**

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# Chapter D1 - Qualitative Variables

### D1 - Question 1

**Question**

The bar plot and the pie chart below show the distribution of pre-existing medical conditions of children involved in a study on the optimal duration of antibiotic use in treatment of tracheitis, which is an upper respiratory infection.

(a) What features are apparent in the bar plot but not in the pie chart?

(b) What features are apparent in the pie chart but not in the bar plot?

(c) Which graph would you prefer to use for displaying these categorical (=qualitative) data?

**Note:** Refer to the original exercise document for graphs.

**Solution**

**Tip : Bar Plot vs. Pie Chart**

A **bar plot** (or a **needle graph**) uses bars to show how many times each category appears. Taller bars mean a category occurs more often. The bars can be arranged from largest to smallest to make comparisons easier.

A **pie chart** shows the same information but as slices of a circle. Bigger slices mean a category is more common. However, it can be harder to compare categories in a pie chart, especially when the slices are similar in size.

**(a) What features are apparent in the bar plot but not in the pie chart?**
The bar plot makes it easy to compare categories because the bars are arranged in order of size. It also shows exact differences between categories because we can see the height of each bar. In contrast, the pie chart does not arrange categories in

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order and makes it harder to compare similar-sized sections.

**Answer:** The bar plot shows clear comparisons between categories by ordering them and making their sizes easier to see.

**(b) What features are apparent in the pie chart but not in the bar plot?**
The pie chart represents all categories as parts of a whole, giving a sense of how much each category contributes to the total. However, it does not provide additional details beyond what the bar plot already shows.

**Answer:** The pie chart visually represents parts of a whole but does not show anything that the bar plot does not.

**(c) Which graph would you prefer to use for displaying these categorical data?**
A bar plot is usually the better choice because it allows for easy comparisons between categories. Since the bars can be ordered by size, it is much easier to see which categories are more or less common. Pie charts can be harder to interpret, especially with many similar-sized categories.

**Answer:** The bar plot is preferred because it makes comparisons easier by showing category sizes clearly.

**D1 - Question 2**

**Question**

In the table below, you find the hair colour for a sample of students.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| blond | brown | black | blond | brown | brown | brown | blond | brown | black |
| black | blond | brown | black | black | black | brown | brown | brown | black |
| black | blond | black | brown | blond | black | blond | brown | brown | brown |

(a) Set up a frequency table with absolute and relative frequencies for the variable *colour*.

(b) Find the mode of the variable *colour*.

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(c) Set up a pie chart for the variable *colour*.

(d) Set up a bar chart for the variable *colour*.

**Solution**

**(a) Frequency table of hair colour**

**Tip : Absolute and Relative Frequency**

The **absolute frequency** tells us how many times a certain value appears in the data. It is just a simple count, denoted as $n_1, n_2, n_3, ...$ for different categories.

The **relative frequency** shows how often a value appears *relative* to the total number of observations. It is calculated as:

$$\text{relative frequency} = \frac{\text{absolute frequency}}{\text{total sample size}}$$

We denote relative frequencies as $\hat{p}_1, \hat{p}_2, \hat{p}_3, ...$. This helps compare categories, even if the total number of observations changes.

- The **random experiment** is selecting a student and recording their hair colour.
- The **variable** is *hair colour*, which is:
  - **Qualitative** because it consists of $k = 3$ categories:

    $$m_1 = \text{black}, \quad m_2 = \text{blond}, \quad m_3 = \text{brown}$$

  - **Nominal** because the categories have no natural order.
- The **sample size** is $n = 30$, meaning we have 30 students. Their hair colours are denoted as $x_1, x_2, \ldots, x_{30}$, using lowercase since these are observed values.

To find the frequencies, we count how many times each hair colour appears:

$$n_1 = 10, \quad n_2 = 7, \quad n_3 = 13$$

Now, we calculate the relative frequencies:

$$\hat{p}_1 = \frac{10}{30} = 0.3333, \quad \hat{p}_2 = \frac{7}{30} = 0.2333, \quad \hat{p}_3 = \frac{13}{30} = 0.4333$$

**Answer:** The frequency table is shown below.

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| Colour | Absolute Frequency | Relative Frequency |
|---|---|---|
| Black | 10 | 0.3333 (or 33.33%) |
| Blond | 7 | 0.2333 (or 23.33%) |
| Brown | 13 | 0.4333 (or 43.33%) |
| **Total** | 30 | 1 (or 100%) |

### (b) Find the mode

**Tip : Mode**

The **mode** is the value that appears most often in the data.

For example, in the dataset: {1, 2, 2, 3, 3, 3, 4}, the mode is **3** because it appears the most times.

The mode is the hair colour that appears most often. From the frequency table, brown appears 13 times, which is more than black (10 times) and blond (7 times).

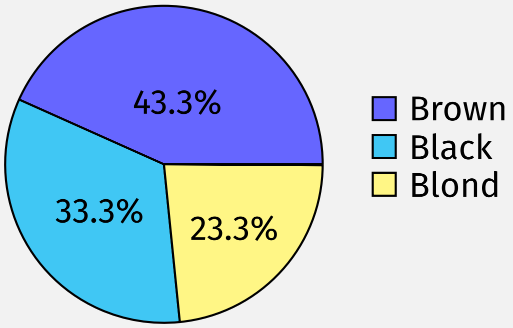
**Answer:** The mode is brown.

### (c) Draw a pie chart

To sketch a pie chart:

1. Draw a circle.
2. Divide it into three slices based on the relative frequencies.
3. The largest slice (43.3%) is for brown, the second largest (33.3%) for black, and the smallest (23.3%) for blond.
4. Label each section with the hair colour and percentage.

**Answer:** Pie chart shown below.

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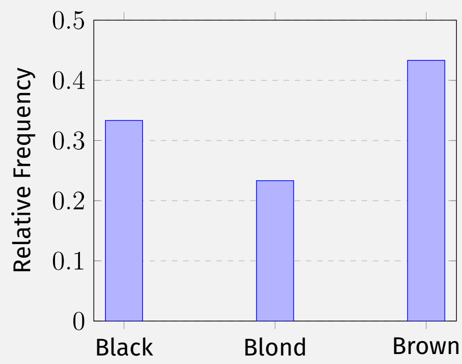
**(c) Draw a bar chart**

To sketch a bar chart:

1. Draw a horizontal axis with category labels (black, blond, brown).
2. Draw a vertical axis labeled ”Relative Frequency.”
3. Draw bars for each category, with heights matching the relative frequencies.

**Answer:** Bar chart displayed below.




**Note** that although bar charts are typically sorted in decreasing order, here we follow the category order introduced in part (a).

### D1 - Question 3

**Question**

In the table below, you find T-shirt sizes for a sample of students, using the following coding:

$$1 = \text{S}, \quad 2 = \text{M}, \quad 3 = \text{L}, \quad 4 = \text{XL}, \quad 5 = \text{XXL}$$

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| | | | | |
|---|---|---|---|---|
| 4 | 2 | 4 | 3 | 1 |
| 2 | 3 | 1 | 2 | 1 |
| 3 | 3 | 3 | 3 | 2 |
| 4 | 4 | 1 | 1 | 2 |
| 3 | 2 | 3 | 1 | 3 |
| 1 | 2 | 1 | 1 | 5 |
| 1 | 4 | 3 | 2 | 2 |
| 3 | 2 | 1 | 2 | 2 |
| 3 | 1 | 4 | 5 | 4 |
| 2 | 3 | 3 | 1 | 3 |

(a) Set up a frequency table for the variable *Size of T-shirts* including absolute and relative frequencies.

(b) Add the cumulative absolute and relative frequencies in the table above.

(c) Create a needle graph and a pie chart for the T-shirt sizes.

**Solution**

**(a) Frequency table of T-shirt sizes**

- The **random experiment** is selecting a student and recording their T-shirt size.
- The **variable** is *T-shirt size*, which is:
  - **Qualitative** because it consists of $k = 5$ categories:

$$m_1 = \text{S}, \quad m_2 = \text{M}, \quad m_3 = \text{L}, \quad m_4 = \text{XL}, \quad m_5 = \text{XXL}$$

  - **Ordinal** because the categories have a natural order (S < M < L < XL < XXL).
- The **sample size** is $n = 50$, meaning we have 50 students. Their T-shirt sizes are denoted as $x_1, x_2, \ldots, x_{50}$, using lowercase since these are observed values.

To find the frequencies, we count how many times each T-shirt size appears:

$$n_1 = 13, \quad n_2 = 12, \quad n_3 = 16, \quad n_4 = 7, \quad n_5 = 2$$

Now, we calculate the relative frequencies:

$$\hat{p}_1 = 0.26, \quad \hat{p}_2 = 0.24, \quad \hat{p}_3 = 0.32, \quad \hat{p}_4 = 0.14, \quad \hat{p}_5 = 0.04$$

**Answer:** The frequency table is shown after part (b).

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**(b) Cumulative frequencies**

**Tip : Cumulative Frequency**

The **cumulative absolute frequency** tells us how many observations are less than or equal to a certain category. It is found by summing the absolute frequencies up to that point.

The **cumulative relative frequency** does the same but for relative frequencies, showing the proportion of data that falls in or below each category. It is denoted as $\hat{F}_i$ and calculated as:

$$\hat{F}_i = \hat{p}_1 + \hat{p}_2 + \cdots + \hat{p}_i$$

Cumulative frequencies are **only meaningful for ordinal variables**, because the order of categories matters. For example, in T-shirt sizes (S, M, L, XL, XXL), cumulative frequency tells us the proportion of students wearing a given size or smaller.

However, for **nominal variables** (e.g., hair colour: black, blond, brown), cumulative frequencies do not make sense since the categories have no natural order.

We sum the absolute and relative frequencies step by step. For example,

$$\hat{F}_3 = \hat{p}_1 + \hat{p}_2 + \hat{p}_3 = 0.26 + 0.24 + 0.32 = 0.82$$
$$\hat{F}_4 = \hat{p}_1 + \hat{p}_2 + \hat{p}_3 + \hat{p}_4 = \hat{F}_3 + \hat{p}_4 = 0.82 + 0.14 = 0.96$$

**Answer:** The frequency table is shown below.

| Size | Code | Abs. Freq. | Rel. Freq. | Cumul. Abs. Freq. | Cumul. Rel. Freq. |
|---|---|---|---|---|---|
| S | 1 | 13 | 0.26 | 13 | 0.26 |
| M | 2 | 12 | 0.24 | 25 | 0.50 |
| L | 3 | 16 | 0.32 | 41 | 0.82 |
| XL | 4 | 7 | 0.14 | 48 | 0.96 |
| XXL | 5 | 2 | 0.04 | 50 | 1.00 |
| **Total** | - | 50 | 1 (or 100%) | - | - |

**(c) Draw a needle graph and a pie chart**

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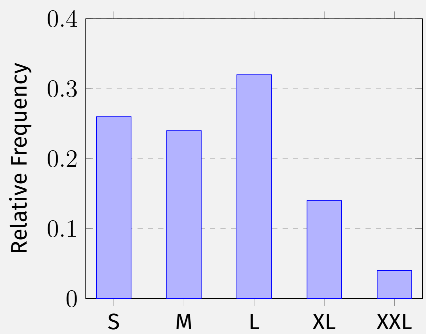
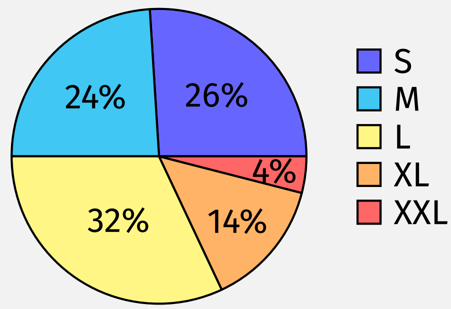
**Tip : Needle Graph**

A **needle graph** and a bar chart are the same thing.

**Answer:** Needle graph and pie chart shown below.




**D1 - Question 4**

**Question**

In a frequency table, absolute frequencies are provided for the different outcomes of the variable *continent* for a sample of BBA students. This variable is a qualitative variable indicating the continent for a randomly selected BBA student.

| **Continent** $m_i$ | $n_i$ | $\hat{p}_i$ |
|---|---|---|
| Europe | 159 | |
| Asia | 54 | |
| N-America | 11 | |
| S-America | 5 | |
| Africa | 26 | |
| Oceania | 1 | |

(a) Determine the sample size.

(b) Add a column with the relative frequencies in the table.

(c) Determine the mode.

(d) Draw a needle graph with absolute frequencies.

(e) Draw a pie chart with relative frequencies.

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**Solution**

**(a) Determine the sample size**

The sample size $n$ is the sum of all absolute frequencies:

$$n = 159 + 54 + 11 + 5 + 26 + 1 = 256$$

**Answer:** The sample size is $n = 256$.

**(b) Frequency table with relative frequencies**

We compute the relative frequencies as follows:

$$\hat{p}_1 = \frac{159}{256} = 0.6211, \quad \hat{p}_2 = \frac{54}{256} = 0.2109, \quad \hat{p}_3 = \frac{11}{256} = 0.0430,$$
$$\hat{p}_4 = \frac{5}{256} = 0.0195, \quad \hat{p}_5 = \frac{26}{256} = 0.1016, \quad \hat{p}_6 = \frac{1}{256} = 0.0039$$

**Answer:** The completed frequency table is shown below.

| **Continent $m_i$** | $n_i$ | $\hat{p}_i$ |
|---|---|---|
| Europe | 159 | 0.6211 |
| Asia | 54 | 0.2109 |
| N-America | 11 | 0.0430 |
| S-America | 5 | 0.0195 |
| Africa | 26 | 0.1016 |
| Oceania | 1 | 0.0039 |
| **Total** | 256 | 1 (or 100%) |

**(c) Determine the mode**

The mode is the continent with the highest absolute frequency. From the table, Europe has the highest count of 159.

**Answer:** The mode is Europe.

**(d) Draw a bar chart with absolute frequencies**

Remember that previously, we created a bar chart using relative frequencies. This time, we follow the same process, but the y-axis (or bar height) represents absolute counts instead of proportions.

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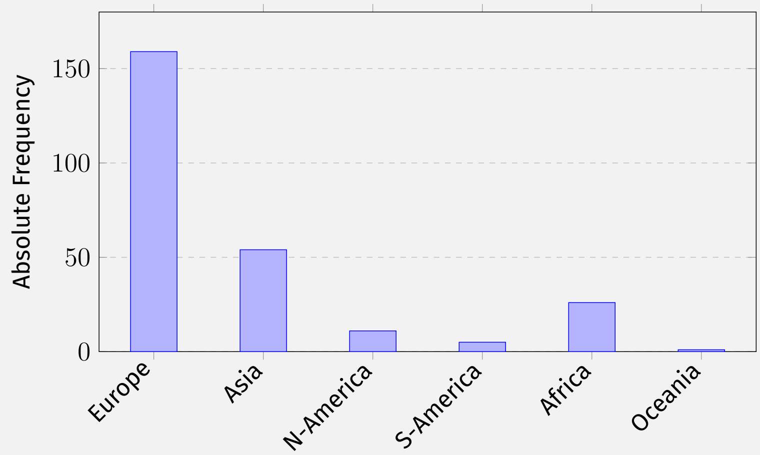
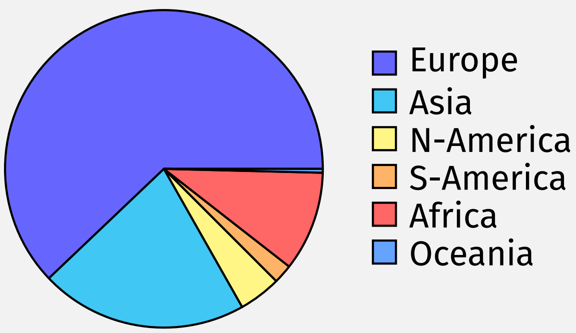
**Answer:** Bar chart shown below.




**(e) Draw a pie chart with relative frequencies.**

**Answer:** The pie chart is displayed below.




### D1 - Question 5

**Question**

For a sample of 519 members of academic staff in the USA, the rank is measured. The following codes are used:

1 = Assistant Professor, 2 = Associate Professor
3 = Full Professor, 4 = Distinguished Professor

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| Code | Abs. Freq. | Rel. Freq. | Cumul. Abs. Freq. | Cumul. Rel. Freq. |
|---|---|---|---|---|
| 1 | | 0.35453 | | |
| 2 | | 0.34104 | | |
| 3 | | 0.25241 | | |
| 4 | | 0.05202 | | |

(a) Complete the frequency table by determining the absolute frequencies, the cumulative absolute frequencies and the cumulative relative frequencies.

(b) Determine the mode.

(c) Draw a needle graph with relative frequencies.

(d) Draw a 100% stacked column chart.

**Solution**

**(a) Complete the frequency table**

- The **random experiment** is selecting a staff member and recording their rank.
- The **variable** is *rank*, which is:
  - **Qualitative** because it consists of $k = 4$ categories:

$$m_1 = \text{Assistant Professor}, \quad m_2 = \text{Associate Professor},$$

$$m_3 = \text{Full Professor}, \quad m_4 = \text{Distinguished Professor}$$

  - **Ordinal** because the categories have a meaningful order from lowest to highest rank.
- The **sample size** is $n = 519$, meaning we have 519 staff members. Their ranks are denoted as $x_1, x_2, \ldots, x_{519}$, using lowercase since these are observed values.

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**Tip : Calculating Absolute Frequency**

The **absolute frequency** for a category is found by multiplying its **relative frequency** by the total sample size:

$$n_i = \hat{p}_i \cdot n$$

Since absolute frequencies represent counts, they must be **rounded to the nearest whole number** to maintain accuracy, as relative frequencies are often given as decimals.
Additionally, the sum of all absolute frequencies must always equal the total sample size.

$$n_1 = 0.35453 \cdot 519 = 183.99987 \approx 184, \quad n_2 = 0.34104 \cdot 519 = 176.99976 \approx 177,$$
$$n_3 = 0.25241 \cdot 519 = 130.99979 \approx 131, \quad n_4 = 0.05202 \cdot 519 = 26.99938 \approx 27$$

To check, the total sum is:

$$184 + 177 + 131 + 27 = 519$$

Since this matches the sample size, the rounding is correct. If the sum had been slightly off due to rounding, we would adjust by increasing or decreasing one of the values to ensure the total remains $n$.

Now, we compute the cumulative absolute and relative frequencies:

$$\begin{aligned} c_1 &= 184 & c_2 &= 184 + 177 = 361, \\ c_3 &= 361 + 131 = 492 & c_4 &= 492 + 27 = 519 \end{aligned}$$

$$\begin{aligned} \hat{F}_1 &= 0.35453 & \hat{F}_2 &= 0.35453 + 0.34104 = 0.69557, \\ \hat{F}_3 &= 0.69557 + 0.25241 = 0.94798 & \hat{F}_4 &= 0.94798 + 0.05202 = 1 \end{aligned}$$

**Answer:** The completed frequency table is shown below.

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| Code | Abs. Freq. | Rel. Freq. | Cumul. Abs. Freq. | Cumul. Rel. Freq. |
|---|---|---|---|---|
| 1 | 184 | 0.35453 | 184 | 0.35453 |
| 2 | 177 | 0.34104 | 361 | 0.69557 |
| 3 | 131 | 0.25241 | 492 | 0.94798 |
| 4 | 27 | 0.05202 | 519 | 1.00000 |
| **Total** | 519 | 1 (or 100%) | - | - |

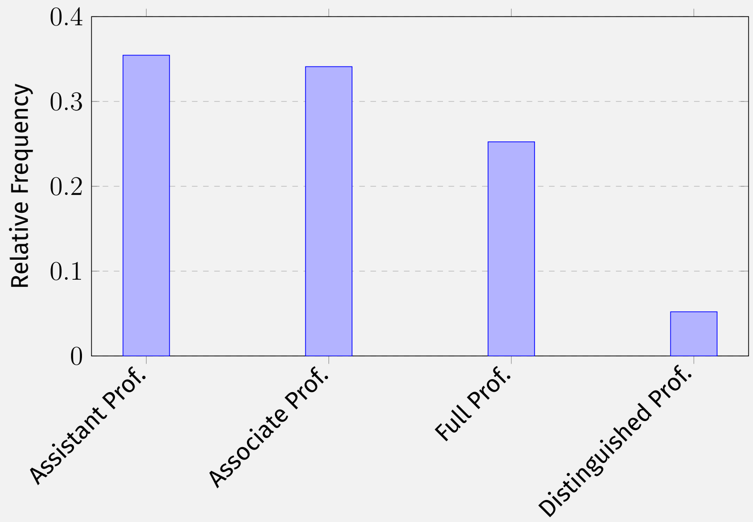
### (b) Determine the mode

The mode is the category with the highest absolute frequency. From the table, Assistant Professor has the highest count of 184.

**Answer:** The mode is Assistant Professor.

### (c) Draw a bar chart with relative frequencies

**Answer:** Bar chart shown below.




### (d) Draw a 100% stacked column chart

**Tip : 100% Stacked Column Chart**

A **100% stacked column chart** represents cumulative relative frequencies using stacked bars. Each section of the bar corresponds to the relative frequency, making it easy to compare category contributions within a whole.

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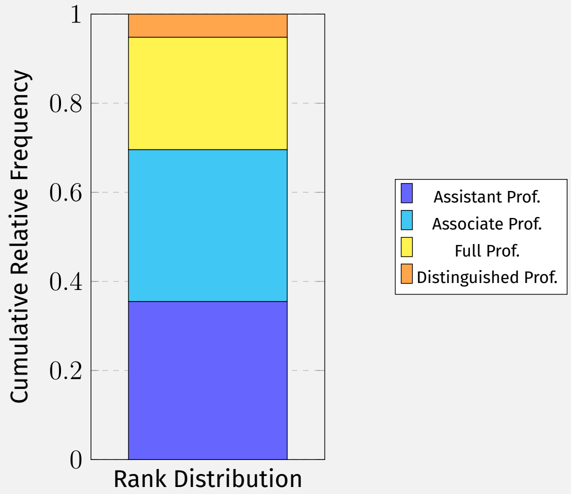
**Answer:** The stacked column chart is shown below.




**D1 - Question 6**

**Question**

For a population of employees, a sample was taken, and the following age distribution was observed. The first group corresponds to ages 18 to 25.

| Age Group | $\hat{p}_i$ |
|---|---|
| 1 | 0.15 |
| 2 | 0.25 |
| 3 | 0.40 |
| 4 | 0.20 |

Now consider the subpopulation of employees **older than 25**. Determine the cumulative relative frequencies for the age groups 2, 3, and 4.

**Solution**

- The **random experiment** is selecting an employee and recording their age group.
- The **variable** is *age group*, which is:
  - **Qualitative** because it consists of $k = 4$ categories:

    $$m_1 = [18,25], \quad m_2, \quad m_3, \quad m_4$$

  - **Ordinal** because the categories have a meaningful order from young to old.

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The subpopulation of interest consists of employees **older than 25**, meaning we exclude group $m_1$ and focus on groups $m_2$, $m_3$, and $m_4$. The new sample size for this subpopulation is:

$$n_2 + n_3 + n_4$$

**Tip : Rescaling Relative Frequencies**

When focusing on a subpopulation, we must **rescale** the relative frequencies so that they sum to 1. This is done by dividing each remaining relative frequency by the sum of the selected groups:

$$\hat{p}_i^{\text{new}} = \frac{n_i}{n_2 + n_3 + n_4}$$

This ensures that the new relative frequencies are properly adjusted within the subpopulation.

However, since we only have relative frequencies and not absolute frequencies, we express this in terms of proportions by dividing everything by the total sample size $n$. This gives:

$$\hat{p}_i^{\text{new}} = \frac{n_i/n}{(n_2 + n_3 + n_4)/n} = \frac{\hat{p}_i}{\hat{p}_2 + \hat{p}_3 + \hat{p}_4}$$

Now, we compute the rescaled relative frequencies:

$$\hat{p}_2^{\text{new}} = \frac{0.25}{0.85} = 0.2941, \quad \hat{p}_3^{\text{new}} = \frac{0.40}{0.85} = 0.4706, \quad \hat{p}_4^{\text{new}} = \frac{0.20}{0.85} = 0.2353$$

To verify, the sum of the new relative frequencies must also be equal to 1:

$$0.2941 + 0.4706 + 0.2353 = 1$$

**Answer:** The new relative frequencies are shown in the table below.

| Age Group | Rel. Freq. |
|---|---|
| 2 | 0.2941 |
| 3 | 0.4706 |
| 4 | 0.2353 |

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